

Takaharu Tanaka

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EDUCATION

University of Washington

Seattle, WA

Master of Science in Electrical and Computer Engineering (combined BS/MS program) Sep. 2027 – Expected June 2028

University of Washington

Seattle, WA

Bachelor of Science in Electrical and Computer Engineering

Sep. 2023 – Expected June 2027

GPA: 3.8 | Relevant Coursework (through Spring 2027): Fundamentals of EE, Design Digital Circuits and Systems, Analog Embedded System, Digital VLSI: Transistors to Gates, Digital VLSI: Gates to System, Fundamentals of Integrated Circuit Technology

EXPERIENCE

Research Assistant

Sep 2025 – Present

EMIT Lab, University of Washington

Seattle, WA

- First-author manuscript on GPU benchmarking of simulated bifurcation, submitted to *IEEE Transactions on Emerging Topics in Computing* (Aug 2026), under review
- Built a PyTorch benchmarking framework for the ballistic simulated bifurcation (SB) solver on an NVIDIA Jetson Orin Nano (MAXCUT and QUBO, 100 to 4500 nodes), sweeping INT8, FP16, FP32, and FP64 precision
- Designed a reproducible measurement methodology (locked GPU clocks, warm-up runs, thermal-throttle guarding, multi-seed statistics) and profiled memory bandwidth and kernel launch overhead

Field Application Engineering Intern

Jul 2026 – Aug 2026

Texas Instruments

Osaka, Japan

- Created a system-level block diagram mapping a customer's application requirements onto a device solution
- Characterized an IC's thermal response from room temperature to 160 °C using an oscilloscope and chamber
- Analyzed a consumer-electronics USB power delivery implementation to estimate current and interface requirements for future designs

Teaching Assistant, Digital Circuits and Systems

Dec 2025 – Mar 2026

University of Washington

Seattle, WA

- Guided a class of 100 students through digital circuit design, debugging, and SystemVerilog, Quartus, and ModelSim workflows in labs and office hours
- Graded assignments and lab reports against design requirements with actionable feedback

PROJECTS

5T OTA Design | Cadence Virtuoso, ADE Assembler, Analog, CMOS

May 2026 – Jun 2026

- Designed a 5-transistor OTA in Cadence Virtuoso on the GPD045 45 nm process, sizing the differential pair, current-mirror load, and 1.5 mA tail bias from hand calculations, then tuning with constrained optimization
- Met all 9 specifications in simulation: 10.12 V/V open-loop gain, 1.501 V output swing, 1.134 GHz unity-gain bandwidth, and 4.5 V/V closed-loop gain in a differential amplifier configuration

AC-DC Boost Converter | Multisim, Power Electronics, Oscilloscope

Nov 2025 – Dec 2025

- Designed and built a 7.5 Vrms to adjustable 10–20 V DC supply: full-wave bridge rectifier, 555-clocked NMOS boost stage, and zener-bounded regulator set by a single potentiometer
- Validated in Multisim, then bench tested at 10 k Ω to 100 k Ω loads, holding ripple to 80 mV at rated load against a 100 mV spec and diagnosing simulation-to-hardware gaps in clock frequency and light-load ripple

5-Stage Pipelined CPU | SystemVerilog, ModelSim, ARM

Sep 2025 – Nov 2025

- Designed and implemented a 5-stage pipelined CPU (IF, ID, EX, MEM, WB) in SystemVerilog with data forwarding and accelerated branch handling to minimize stalls and hazards
- Verified functional correctness at gate level through cycle-accurate simulation in ModelSim

TECHNICAL SKILLS

Analog & Hardware: Cadence Virtuoso (ADE Assembler), LTSpice, Multisim, KiCad, PCB design/testing, oscilloscope, thermal chamber

Digital Design: SystemVerilog, Quartus Prime, ModelSim, FPGA (DE1-SoC), FSMs, ASMD, testbenches

Programming & Tools: Python, PyTorch, CUDA, NVIDIA Jetson, C, MATLAB, Git, Linux